Project Design Phase

Solution Architecture

Date	28-10-2026
Team ID	SWTID-2026-7875
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value

1. Executive Summary & Architecture Overview

This Solution Architecture document defines the structural framework, security models, and data governance policies implemented for script-controlled Access Control Lists (ACLs) within ServiceNow . The architecture ensures granular, database-level record segregation by evaluating user role memberships alongside field-level dynamic attributes.

2. Core Architectural Components

Identity & Role Management Subsystem

- Provisioned dedicated user identity ('EEE User', eeeuser@gmail.com).
- Created operational custom roles bb1 (Read), bb2 (Create), bb3 (Write), and bb4 (Delete).
- Assigned all custom operational privileges directly to the user profile.

Data Schema & Storage Layer

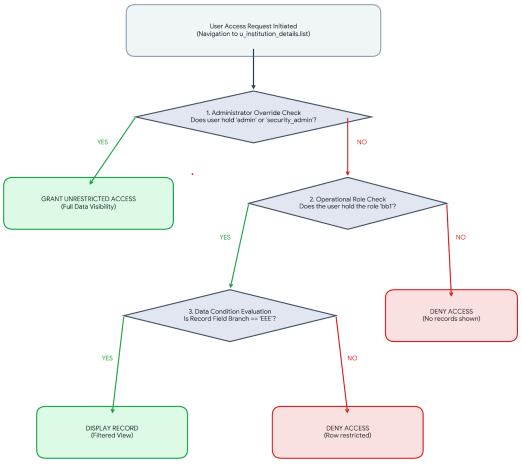
- Established custom table u_institution_details extending base dictionary definitions [cite: 15].
- Configured Student Roll Number with auto-number sequence generation (getNextObjNumberPadded().
- Created user reference fields (sys_user) for Student Name and Faculty Name, alongside Branch choice options (EEE, ECE, CSE).

Security & Access Control Engine

- Read ACL: Restricts visibility using role bb1 and data condition Branch == 'EEE' combined with script logic.

- Create ACL: Governs record creation rights and button rendering via role bb2.
- Write ACL: Enforces field-level and record modification authorization via role bb3.
- Delete ACL: Secures record deletion actions through role bb4 .

3. End-to-End Solution Architecture Flowchart



4. Solution Security & Governance Matrix

ACL Operation	Target Element	Required Role	Enforcement Scope & Mechanism
Read	u_institution_details]	bb1	Restricts viewing strictly to Branch == 'EEE' records via script evaluation
Create	u_institution_details	bb2	Governs record creation authorization and renders the 'New' button
Write	u_institution_details	bb3	Enforces field modification and record update rights across forms
Delete	u_institution_details]	bb4	Secures single and bulk record deletion actions from list view